

Consensus in Multi-Agent Systems

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Abstract—Many cooperative behaviors of multi-agent teams emerge from *local* interactions among the agents, where an agent interacts with a few “adjacent” teammates, but has no information about the remaining agents. For instance, the self-organization of many biological populations including swarms of insects, flocks of birds and schools of fish, are based on such local interaction rules: the motion and decisions of an individual agent are determined by the behavior of its *nearest neighbors* in the population. A special case of multi-agent coordination is *consensus*, that is, agreement of agents on some quantity of interest or, more generally, full or partial synchronization of their state trajectories. Establishing consensus is a “benchmark” problem in multi-agent systems study, which allows to reveal the main principles of multi-agent coordination and, in particular, the role of the system’s interaction graph (or topology). Consensus lies in the heart of many natural phenomena (e.g. synchronous oscillation of neural cells, which maintains a stable heart rhythm) and engineering designs (e.g. attitude synchronization of satellites). In this article, we give a brief review of distributed consensus algorithms, focusing on the basic ideas and relevant mathematical theory, in particular, graph-theoretic methods.

I. INTRODUCTION

The term “consensus” originates from the Latin word *consensio* (unanimity, harmony, coincidence, sympathy). In multi-agent systems, consensus means the agents’ agreement on some quantity of interest or convergence to a common decision. A more general meaning of this term is the full or partial synchronism between the agents’ state trajectories, demonstrated e.g. by the collective motion of herds of animals.

The simplest consensus algorithm, studied in the mathematical literature, seems to be the *cyclic pursuit* strategy [1], [2], dating back to the widely known “Three Dog Problem” [3], [4] and its extension, known as the “problem of N bugs” [1]. In its original version, this problem deals with a group of N mobile agents (“bugs”), which start at the vertices of some regular N -gon and cyclically chase each other at a constant speed (Fig. 1) until they meet at the center of the N -gon. The main concern of the N -bug problem is to find the agents’ trajectories, or *pursuit curves*. Although this problem may seem artificial, it prominently illustrates the idea of distributed consensus algorithms: the global agreement between the agents, that is, their rendezvous at a known point, emerges from their *local* interactions: each agent uses only the relative position of its “predecessor” and does not need any global information about the system (such as e.g. the number of bugs N). Similar cyclic formations arise in the models of insect swarms [1] and cooperative control of mobile robots [5].

The rigorous study of consensus algorithms originates from two independent research areas, extensively studied since 1950s. The first deals with mathematical modeling of social processes, in particular, opinion evolution in social groups. A simple model of opinion dynamics was introduced by

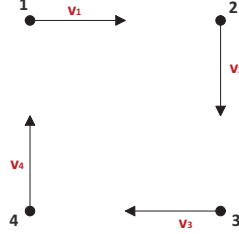


Fig. 1: The cyclic pursuit dynamics of $N = 4$ bugs: each bug pursues the neighbor in cyclic order at a fixed speed $|v_i| = v$.

French [6] and examined by Harary [7] who obtained graph-theoretic conditions for the consensus (“unanimity”) of opinions in the French model. The model from [6] deals with an iterative procedure of opinion formation; at each stage of this procedure agents simultaneously average their opinions with opinions of several “neighbors”, where the neighboring relations (“social ties”) are described by a directed interaction graph. The focus of French’s work was to study the influence of an individual’s initial opinion on the final consensus opinion of the group, treated as his/her capability to control the group’s behavior, or *social power* [6]. The French model lies in the heart of more realistic models of opinion evolution, where the agents’ opinions do not always converge to agreement, but can form several *clusters* [8]. The most typical reasons for the opinion cleavage are *homophily* (an individual readily adopts the opinions of like-minded agents, taking the others’ opinions with discretion or simply ignoring them) [9]–[11], *stubbornness* [12]–[14] (some agents are “attached” to their initial opinions, or *prejudices*) and *antagonism* (distrust, competition) between some pairs of agents [15]–[19].

Following a parallel line of research, the problem of consensus of *subjective probabilities* was considered in applied statistics and management science [20], [21]. Each of N experts has an individual estimate of some uncertain probability distribution; based on these “subjective probabilities”, the experts have to elaborate a *collective* estimate of the unknown distribution. The simplest approach, called *opinion pooling* [21], [22], is to take a weighted average of the individual distributions. Developing this idea, DeGroot [23] proposed an *iterative* procedure of opinion pooling, extending the French model [6] and converging to the *consensus* probability distribution.

The French-DeGroot averaging procedure, along with its extensions, has inspired numerous algorithms for multi-agent consensus. The theory of such algorithms is rapidly developing; its applications include, but are not limited to, efficient algorithms for distributed computing [24], [25], optimization [26], [27] and filtering [28], traffic flow modeling [29], [30], opinion dynamics in social networks [11], [13], [31], [32] and cooperative control of autonomous vehicles [33]–[35].

In this article, we give a brief review of basic distributed algorithms (or *protocols*) for multi-agent consensus algorithms and mathematical tools used for their examination. The text is organized as follows. Section II introduces several concepts from graph theory, used in the analysis of multi-agent systems. In Section III we introduce the French-DeGroot averaging procedure and its continuous-time counterpart, which establish consensus among the agents with the simplest first order dynamics and can be examined by tools from linear algebra and matrix analysis. Section IV gives a review of more sophisticated protocols, providing consensus over time-varying interaction graphs. In this section the effects of delays are also discussed. In Section V, consensus problems for agents with general dynamics are addressed. Section VI discusses some important applications of consensus algorithms.

II. PRELIMINARIES: GRAPH THEORY

In this section, we summarize some graph-theoretic concepts for analysis of multi-agent systems [34]–[36].

A. Graphs and their connectivity properties

A graph consists of *nodes* (or vertices) and *arcs* (or edges). A node can be represented by a dot in the plane, while an arc is a directed line or curve, connecting a dot to itself or another dot (Fig. 2). Mathematically, the nodes are the elements of some finite set $\mathcal{V}$, whereas the arcs are their ordered pairs $e = (v, v') \in \mathcal{V} \times \mathcal{V}$, constituting the set of arcs $\mathcal{E} = \{e\}$.

Definition 1: (Graph, nodes and arcs) A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is a pair of two finite sets $\mathcal{V}$ and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. The set $\mathcal{V}$ is referred to as the set of *nodes* and $\mathcal{E}$ is called the set of *arcs*. The arc $e = (v, v')$ is said to *connect* the node v to the node v' ; an arc $e = (v, v)$ connecting a node to itself is called a *self-loop*. The graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is called *undirected* if all connections are mutual, i.e. for any arc $e = (v, v') \in \mathcal{E}$ the inverse arc $e' = (v', v)$ exists; otherwise, the graph is *directed*.

An undirected graph is often defined [37] as a collection of nodes and *bidirectional* arcs, defined as unordered pairs of nodes $\{v, v'\}$. However, it is convenient to “split” such an arc into the pair of opposite arcs (v, v') and (v', v) , treating thus undirected and directed graphs in a unified way. To illustrate Definition 1, consider two graphs with $N = 4$ nodes in Fig. 2. The graph in Fig. 2a is directed (e.g. the arc $(1, 3)$ has no inverse) and has self-loops at each node; the graph in Fig. 2b is undirected and has no self-loops. Such a graph, where any two distinct nodes are connected, is said to be *complete*.

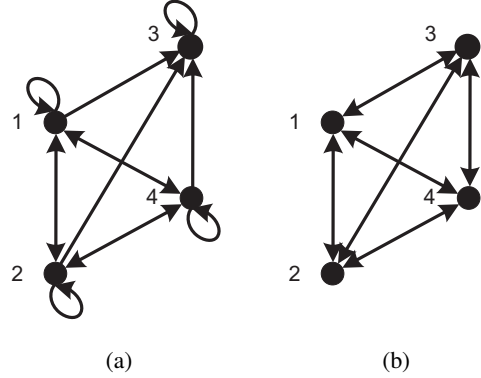
Choosing several nodes from a graph and all arcs between these nodes, one gets a *subgraph*, formally defined as follows.

Definition 2: (Subgraph) Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, its *subgraphs* are graphs $\mathcal{G}' = (\mathcal{V}', \mathcal{E}')$, where $\mathcal{V}' \subseteq \mathcal{V}$ and $\mathcal{E}' = \mathcal{E} \cap (\mathcal{V}' \times \mathcal{V}')$.

For instance, each single node is a subgraph of the graph in Fig. 2b, and any single node together with its self-loop is a subgraph of the graph in Fig. 2a.

Definition 3: (Walks and paths in a graph) A sequence of nodes $v_0, v_1, \dots, v_k \in \mathcal{V}$ is said to be a *walk* of length $k \geq 1$ from v_0 to v_k in the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, if any two consecutive nodes are connected $(v_0, v_1), \dots, (v_{k-1}, v_k) \in \mathcal{E}$.

Fig. 2: A directed and an undirected graphs with $N = 4$ nodes



A walk starting and ending at the same node $v_0 = v_k$ is called a *cycle*; a walk without repeating nodes ($v_i \neq v_j$ when $i \neq j$) is referred to as a *path*.

Definition 4: (Rootedness and connectivity) A node $v_0 \in \mathcal{V}$, from where walks to all other nodes exist, is referred to as a *root* of the graph. The graph is *rooted* or *quasi-strongly connected* if it has at least one root. The graph is called *strongly connected* if any of its nodes is a root.

The rootedness of a graph is equivalent to the existence of a *directed spanning tree* in the graph [34]. For undirected graphs, quasi-strong and strong connectivity are equivalent; such a graph is said to be *connected*. The complete graph in Fig. 2b is connected; the directed graph in Fig. 2a has three root nodes 1, 2, 4 yet is not strongly connected, since node 3 is not a root. When the graph is not strongly connected, its nodes are divided between several strongly connected *components*.

Definition 5: A strongly connected component of the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is a strongly connected subgraph $\mathcal{G}' = (\mathcal{V}', \mathcal{E}')$, which is *maximal* in the sense that any larger subgraph $\mathcal{G}'' = (\mathcal{V}'', \mathcal{E}'')$, where $\mathcal{V}' \subsetneq \mathcal{V}''$, is not strongly connected.

Any node of the graph belongs to the only strongly connected component. For an undirected graph, between different (strongly) connected components there are no arcs; in the directed graph case, different components may be connected. For instance, the graph in Fig. 2a has two strongly connected components, whose node sets are $\{1, 2, 4\}$ and $\{3\}$; the former component is connected to the latter one by three arcs $(1, 3)$, $(2, 3)$ and $(4, 3)$.

Remark 1: If the graph is rooted, then all of its roots belong to the same strongly connected component and any node from this component is a root.

Definition 6: The greatest common divisor of all cycles in a graph is called the graph's *period*; the period of a graph without cycles, by definition, equals 1. The graph of period 1 is said to be *aperiodic*.

For instance, any graph having at least one self-loop is aperiodic. Connecting the nodes in cyclic order $1 \mapsto 2 \mapsto \dots \mapsto N \mapsto 1$, one obtains a graph of period N .

B. Weighted graphs, adjacency matrices and Laplacians

Henceforth we consider graphs $(\mathcal{V}, \mathcal{E})$, whose nodes are indexed 1 through N , that is, $\mathcal{V} = \{1, 2, \dots, N\}$. A nonnegative

matrix $A = (a_{ij})_{i,j=1}^N$ (where $a_{ij} \geq 0$) is called the *weighted adjacency matrix* if $a_{ij} > 0$ when $(i, j) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. If the graph has an arc $e = (i, j)$ then $a_{ij} > 0$ may be treated as a *weight* or *value* of the arc e . A graph whose arcs are assigned with (positive) weights is said to be *weighted*. The rigorous definition is as follows.

Definition 7: (Weighted graph) A *weighted graph* is a triple $(\mathcal{V}, \mathcal{E}, A)$, where $(\mathcal{V}, \mathcal{E})$ is a graph and A is its weighted adjacency matrix. Since $\mathcal{E} = \{(i, j) : a_{ij} > 0\}$ is uniquely defined by the adjacency matrix, we henceforth denote such a weighted graph $\mathcal{G}[A]$. A weighted graph is *undirected* if $A = A^\top$ (this implies the undirectedness of the graph $(\mathcal{V}, \mathcal{E})$).

The structure of walks (and hence the connectivity properties) of a weighted graph depends only on its set of arcs $\mathcal{E}$ but not on a specific adjacency matrix A . For any choice of the adjacency matrix A , a walk of length $n \geq 1$ from node i to node j exists if and only if $(A^n)_{ij} > 0$. Since the *shortest* walk between any two nodes has the length $n \leq N - 1$, the graph is strongly connected if and only if the matrix $I + A + \dots + A^{N-1}$ is strictly positive.

Definition 8: (Laplacian) For a weighted graph $\mathcal{G}[A]$ without self-loops ($a_{jj} = 0$), the *Laplacian* matrix is defined by

$$L[A] \triangleq (l_{ij}), \quad l_{ij} \triangleq \begin{cases} -a_{ij}, & i \neq j \\ \sum_{k=1}^N a_{ik}, & i = j. \end{cases} \quad (1)$$

The Laplacian matrix always has an eigenvalue at 0, corresponding to the eigenvector $\mathbf{1}_N = (1, 1, \dots, 1)^\top \in \mathbb{R}^N$. Its algebraic multiplicity (the multiplicity as a root of the characteristic equation $\det(\lambda I - L) = 0$) and geometric multiplicity (the dimension $\dim \ker L$) coincide [38]; the zero eigenvalue is *simple* (have multiplicity 1) if and only the graph is rooted [34], [38]. For an undirected weighted graph, the Laplacian matrix is symmetric and its second eigenvalue (in the ascending order) $\lambda_2(L)$ is called *algebraic connectivity* of the graph; $\lambda_2(L) > 0$ if and only the graph is connected.

Definition 9: (Balanced graphs) A weighted graph $\mathcal{G}[A]$ is called *balanced* if $\sum_{j=1}^N a_{ij} = \sum_{j=1}^N a_{ji}$ for any i .

The balance condition is similar in spirit to the well-known Kirchhoff law: the inflow of electric charge at each node of an electric circuit is equal to the outflow of the charge. Obviously, undirected weighted graphs are balanced. It can be shown that, similarly to undirected graphs, between different strongly connected components of a balanced graph, there are no connecting arcs (see e.g. [39, Lemma 1]). Thus a balanced graph is rooted if and only if it is strongly connected.

III. ITERATIVE OPINION POOLING AND ITS EXTENSIONS

In this section, we consider a simple consensus algorithm, known as the French-DeGroot model [13] and its continuous-time counterpart. The special case of this algorithm was introduced in [6] in the context of social network analysis. A more general iterative procedure for the collective decision making was proposed by DeGroot in [23] under the name of *iterative opinion pooling* and, independently, by Lehrer [40] and Wagner [41]. The French-DeGroot model may also be treated as a process of non-Bayesian *learning* [42], [43].

A. The French-DeGroot model

The French-DeGroot model deals with a team of N agents, aiming to agree on some quantity of interest, e.g. the probability of some outcome in the random experiment [23]. Each agent has its own estimate of this quantity $x_i \in \mathbb{R}$, often called *opinion*; the model describes a discrete-time evolution of the agents' opinions $x_i = x_i(t)$. At each stage $t = 0, 1, 2, \dots$ of this iteration an agent simultaneously updates its opinion to the weighted average of the teammates' opinions

$$x_i(t+1) = \sum_{j=1}^N w_{ij} x_j(t), \quad \forall i = 1, \dots, N \quad \forall t = 0, 1, \dots \quad (2)$$

Here $w_{ij} \geq 0$ is the fixed weight, assigned by the i th agent to the j th agent's opinion, and it is assumed that $\sum_{j=1}^N w_{ij} = 1$, i.e. the correspondent square matrix $W = (w_{ij})$ is *stochastic*.

Definition 10: (Stochastic matrix) A matrix $W = (w_{ij})$ with non-negative entries whose rows sum to 1, that is, $\sum_{j=1}^N w_{ij} = 1$, is referred to as a (*row*-) *stochastic* matrix.

The dynamics (2) can be rewritten in the vector form as

$$x(t+1) = Wx(t), \quad x(t) \triangleq (x_1(t), \dots, x_N(t))^\top \in \mathbb{R}^N. \quad (3)$$

More generally, one may assume $x_i(t)$ to be row vectors $x_i(t) = [x_{i1}(t), \dots, x_{im}(t)]$ (e.g. the probability distribution [23]); stacking them one onto each other, one obtains a matrix $X(t)$ of size $N \times m$ that evolves similarly to (3)

$$X(t+1) = WX(t), \quad X(t) = \begin{bmatrix} x_{11}(t) & \cdots & x_{1m}(t) \\ \vdots & \ddots & \vdots \\ x_{N1}(t) & \cdots & x_{Nm}(t) \end{bmatrix}. \quad (4)$$

In the paper [23] the procedure (2) was called *iterative opinion pooling*. The main concern of [23] was to find conditions, providing *consensus* of the agents' opinions.

Definition 11: (Consensus) The algorithm (2) establishes *consensus* among the agents if for arbitrary initial condition their opinions converge to a common limit

$$\lim_{t \rightarrow \infty} x_1(t) = \dots = \lim_{t \rightarrow \infty} x_N(t). \quad (5)$$

It can be shown [23], [35] that the algorithm (2) establishes consensus if and only if the powers W^k converge as $k \rightarrow \infty$ to a stochastic matrix W_* with identical rows

$$\lim_{k \rightarrow \infty} W^k = W_* = \mathbf{1}_N p^\top, \quad p = (p_1, \dots, p_N)^\top \in \mathbb{R}^N. \quad (6)$$

Here $p_i \geq 0$ and $p^\top \mathbf{1}_N = \sum_{j=1}^N p_j = 1$. The condition (6) implies that the opinions converge to the consensus value

$$\lim_{t \rightarrow \infty} x_j(t) = \sum_{i=1}^N p_i x_i(0) \quad \forall j = 1, \dots, N, \quad \forall x(0). \quad (7)$$

The coefficient $p_i \geq 0$ thus indicates the influence of the i th agent's initial opinion on the consensus opinion of the group and can be treated as a measure of the agent's *social power* [6].

Definition 12: (SIA matrix) A stochastic matrix W , satisfying (6), is said to be *SIA* (stochastic indecomposable aperiodic) [34], [44].

SIA matrices have been long studied in matrix analysis and Markov chain theory [45]–[47]. Any stochastic $N \times N$ matrix

may be considered as the matrix of *transition probabilities* of some Markov chain with N states [45]. The SIA condition (6) implies that the Markov chain “forgets” its history and converges to the unique stationary distribution $(p_1, \dots, p_N)$, that is, the probability of the chain’s being in the state i on the k th step converges to p_i as $k \rightarrow \infty$. Such Markov chains are called *regular* [45], and SIA matrices are also referred to as *regular* [45] or *fully regular* [47] stochastic matrices.

Given a stochastic matrix W , the SIA condition can be tested by computing its spectrum [34], [47]: the matrix is SIA if and only if $\lambda_1 = 1$ is the only eigenvalue on the unit circle $|\lambda| = 1$, while the remaining $N - 1$ eigenvalues $\lambda_2, \dots, \lambda_N$ belong to the open unit disk $|\lambda_j| < 1$. In particular, the vector p from (6) is the only *left* eigenvector of W , such that $p^\top W = p^\top$ and $p^\top \mathbf{1}_N = 1$. A sufficient condition for being SIA is *primitivity* [23], [45], [46]: W^n is a strictly positive matrix for some $n \geq 1$. More convenient, however, are *graphical* criteria that employ the interaction graph (or *topology*) associated with the matrix of the multi-agent system. Unlike algebraic approaches, these criteria can be extended to the case of time-varying interactions. We start with the following important definition.

Definition 13: (Interaction graph) A weighted *interaction graph* of the multi-agent team (2) is the graph $\mathcal{G}(W^\top)$.

Following the tradition, established in multi-agent consensus theory [34], [35], [48], [49], we associate the interaction graph with the *transposed* matrix $W^\top$, that is, if $w_{ij} > 0$ then the node j is connected to the node i . The arcs represent the information flow between the agents: when $w_{ij} > 0$, the i th agent uses information about the j th agent’s opinion $x_j(t)$ at each stage of the opinion iteration. The following theorem gives a criterion for consensus of the opinions (5).

Theorem 1: The algorithm (2) establishes consensus, that is, W is SIA, if and only if the interaction graph is *rooted* and the strongly connected component of its roots is aperiodic.

Theorem 1 is a reformulation of the well-known criterion of Markov chain’s regularity, see e.g. [45, Theorem 4.7] (the strongly connected component, consisting of root nodes, corresponds to the class of mutually connected “essential” or “recurrent” states of the chain). Theorem 6 provides consensus when $\mathcal{G} = \mathcal{G}(W^\top)$ is rooted and all its nodes have self-loops [34, Corollary 2.18 and Lemma 2.19]. Another case where consensus is established is a strongly connected aperiodic interaction graph; the latter condition in fact is equivalent to the primitivity of the matrix W [46, Theorem 8.5.3].

As will be discussed later in Subsect. III-D, the graph’s rootedness is a fundamental requirement for any coordinated behavior of the agents. If the aperiodicity assumption is violated, the agents’ opinions usually do not even converge. To illustrate this, consider a team of $N = 2$ agents, obeying

$$x_1(k+1) = x_2(k), \quad x_2(k+1) = x_1(k).$$

Obviously, one has $x_1(2k+1) = x_2(2k) = x_2(0)$ and $x_2(2k+1) = x_1(2k) = x_1(0)$ for any $k \geq 0$; in the case where $x_1(0) \neq x_2(0)$ the agents’ opinions oscillate. In fact, if the strongly connected component of the root nodes is periodic, then W has an eigenvalue on the unit circle $\{\lambda \in \mathbb{C} : |\lambda| = 1\}$, which is different from 1, so that W^k has no limit [47].

B. Continuous-time consensus dynamics

A continuous-time counterpart of the discrete-time dynamics (2), also providing the contraction property, is as follows¹

$$\dot{x}_i(t) = \sum_{j=1}^N a_{ij}[x_j(t) - x_i(t)], \quad \forall i = 1, \dots, N \quad \forall t \geq 0. \quad (8)$$

Here $A = (a_{ij})$ is an *arbitrary* non-negative matrix (not necessarily stochastic); without loss of generality, we may assume that $a_{ii} = 0 \forall i$. Although the algorithm (8) is known since 1960s [50], necessary and sufficient conditions for its convergence have been obtained only recently [34], [36], [51] and are based on the properties of Laplacian matrices.

Introducing the column vector of opinions $x(t) = (x_1(t), \dots, x_N(t))^\top$, the algorithm (8) can be rewritten as

$$\dot{x}(t) = -L[A]x(t) \quad \forall t \geq 0, \quad (9)$$

where $L[A]$ is the Laplacian (1). In the multidimensional case the algorithm (8), similarly to (4), can be rewritten as

$$\dot{X}(t) = -L[A]X(t), \quad X(t) = \begin{bmatrix} x_{11}(t) & \cdots & x_{1m}(t) \\ \vdots & \ddots & \vdots \\ x_{N1}(t) & \cdots & x_{Nm}(t) \end{bmatrix}.$$

A necessary and sufficient condition for convergence of the solutions of (8) to consensus (5) is equivalent to the condition

$$\lim_{t \rightarrow +\infty} \exp(-L[A]t) = \mathbf{1}_N p^\top, \quad p \in \mathbb{R}^N. \quad (10)$$

If the latter condition holds, the consensus opinion is given by (7), similarly to the discrete-time case.

The matrix $(-L[A])$ has an eigenvalue at 0, corresponding to the eigenvector $\mathbf{1}_N$. The algebraic and geometric multiplicities of the zero eigenvalue coincide [38], and there are no other eigenvalues in the unstable half-plane $\{\lambda \in \mathbb{C} : \operatorname{Re} \lambda \geq 0\}$. Hence (10) holds if and only if 0 is a *simple* eigenvalue of $L[A]$; in this case the vector p is the only left eigenvector, such that $p^\top L[A] = 0$ and $p^\top \mathbf{1}_N = 1$; it can be shown that $p_i \geq 0 \forall i$. The latter algebraic conditions can be reformulated in terms of the interaction graph [34], [36], [38], defined, similarly to the protocol (2), as the weighted graph $\mathcal{G}[A^\top]$.

Theorem 2: The algorithm (8) establishes consensus (5) if and only if the interaction graph $\mathcal{G}[A^\top]$ is *rooted*.

Using the continuous-time consensus algorithm (8), the graph periodicity is not needed for consensus. Unlike the DeGroot algorithm, in the continuous case opinions $x_i(t)$ have finite limits as $t \rightarrow \infty$ for any matrix A (see e.g. [52]), although these limits may be different.

C. Special types of consensus

As was discussed, the consensus value (7) is determined by the left eigenvector of the weight matrix W in (2) or the Laplacian $L[A]$ in (8). The computation of this eigenvector for a large-scale system may be a non-trivial problem. However, under special choices of the interaction graph the consensus value can be easily found. In this subsection, we consider

¹If $x(t)$ is a function of time, then $\dot{x}(t)$ stands for its derivative $\dot{x}(t) = \frac{d}{dt}x(t)$. The second derivative is denoted by $\ddot{x}(t) = \frac{d^2}{dt^2}x(t)$.

two special types of consensus, referred to as the *average consensus* and the *leader-following consensus*.

Definition 14: The algorithm (2) or (8) establishes *average consensus* if (7) holds with $p_1 = \dots = p_N = 1/N$, i.e.

$$\lim_{t \rightarrow \infty} x_i(t) = \frac{x_1(0) + \dots + x_N(0)}{N} \quad \forall i = 1, \dots, N. \quad (11)$$

Recalling that p is the left eigenvector of the matrix W (respectively, the Laplacian $L[A]$), corresponding to the eigenvalue 1 (respectively, 0), it can be easily shown that average consensus is established if and only if (6) (respectively, (10)) holds and the interaction graph is *balanced*. Since rooted balanced graphs are strongly connected, Theorems 1 and 2 imply the following important criterion [34], [51], [53].

Theorem 3: The protocol (2) establishes average consensus if and only if the interaction graph $\mathcal{G}[W^\top]$ is balanced, strongly connected and aperiodic. The protocol (8) establishes average consensus if and only if the interaction graph $\mathcal{G}[A^\top]$ is balanced and strongly connected.

Note that in discrete-time, average consensus requires W to be *doubly stochastic* [46]: $\sum_j w_{ij} = \sum_j w_{ji} = 1 \forall i$, that is, both rows and columns of the matrix W sum to 1.

Another special situation is the consensus with a *leader*, where the consensus value of the whole group is fully determined by the initial opinion of one dedicated agent.

Definition 15: The algorithm (2) or (8) establishes *consensus with leader i* if (7) holds with $p_i = 1$ and $p_j = 0 \forall j \neq i$, that is, all the opinions converge to $x_i(0)$.

Since p is the left eigenvector of W (respectively, $L[A]$), consensus with leader i implies that $w_{ii} = 1$ and $w_{ij} = 0 \forall j \neq i$ (respectively, $a_{ij} = 0 \forall j$). In other words, the leader is not influenced by any other agent and keeps its opinion constant $x_i(t) \equiv x_i(0)$. Theorems 1 and 2 imply the following criterion for consensus with leader.

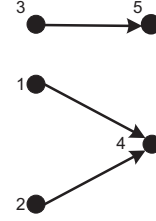
Theorem 4: The protocol (2) (respectively, (8)) establishes consensus with leader i if and only if node i is (the only) root of the interaction graph and has no incoming arcs.

Consensus with a leader in the sense of Definition 15 implies that the leader's state remains unchanged $x_i(t) \equiv x_i(0)$. A more interesting situation is the consensus with a dynamic leader, whose trajectory $r(t) = x_i(t)$ may be considered as a reference to follow or an external control input to the system. Continuous-time algorithms for *reference tracking* consensus, providing that $|x_j(t) - r(t)| \xrightarrow{t \rightarrow \infty} 0$ for any j , can be obtained by a modification of (8), see e.g. [34, Chapter 3], [54].

D. The necessity of the graph's rootedness

As one may notice, the assumption on the graph's rootedness appears to be necessary in both discrete-time and continuous-time cases. An elegant explanation of its necessity was suggested in [49] (Theorem 5). If the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ has no roots, then two non-empty disjoint subsets of nodes $\mathcal{V}_1, \mathcal{V}_2 \subseteq \mathcal{V}$ exist, having no incoming arcs (that is, if $i \in \mathcal{V}_s$ and $j \notin \mathcal{V}_s$ then $(j, i) \notin \mathcal{E}$). In particular, the corresponding two groups of agents are totally independent from each other and thus cannot reach consensus or any other cooperative behavior, no matter which interaction algorithm is used.

Fig. 3: A network with 3 independent “leaders”



An important situation where the interaction graph has no roots is the presence of several static or dynamics leaders, that are mutually independent but influence the remaining agents. Consider, for instance, the team of $N = 5$ agents (Fig. 3), whose scalar opinions $x_i(t)$ obey the equations

$$\begin{aligned} \dot{x}_1(t) &= \dot{x}_2(t) = \dot{x}_3(t) = 0, \\ \dot{x}_4(t) &= a(x_1(t) - x_4(t)) + b(x_2(t) - x_4(t)) \\ \dot{x}_5(t) &= c(x_3(t) - x_5(t)). \end{aligned} \quad (12)$$

Here $a, b, c > 0$ are constant coefficients. Obviously, the “leaders” 1, 2, 3 are attached to their initial opinions $x_1(0), x_2(0), x_3(0)$ (“stubborn”). Unlike them, the “open-minded” agents 4 and 5 “forget” their initial opinions so that

$$x_4(t) \xrightarrow{t \rightarrow \infty} \frac{a}{a+b} x_1(0) + \frac{b}{a+b} x_2(0), \quad x_5(t) \xrightarrow{t \rightarrow \infty} x_3(0).$$

Although such a protocol cannot provide consensus, the following weaker *containment* property is ensured: the limiting opinions are contained by the convex hull, spanned by the static opinions of the leaders. In the case where $x_1(0) = x_2(0) = x_3(0) = \xi \in \mathbb{R}$, this convex hull collapses to a singleton and thus all the opinions agree $x_i(t) \xrightarrow{t \rightarrow \infty} \xi$. The protocol (12) is a special case of *containment control* algorithm [35], providing the convergence of all agents' states into the convex hull, spanned by several static or dynamic leaders. Analogous discrete-time protocols with “stubborn” leaders naturally arise in the opinion dynamics models [13], [14], [55]. The properties of “consensus” algorithms over general graphs without roots have been studied in the recent papers [52], [56], revealing the new deep relations between the steady values of the agents' opinions and spectral properties of the interaction graphs.

E. Extensions: second-order consensus protocols

The results of Theorem 1 and 2 imply the convergence of more complicated consensus algorithms, described by differential equations of second and higher orders. In this subsection we deal with the following second-order continuous-time algorithm, arising e.g. in *rendezvous* problems (see Section VI)

$$\ddot{x}_i = -\rho_i \dot{x}_i + \sum_{j=1}^N a_{ij} [(x_j(t) - x_i(t)) + \gamma_i (\dot{x}_j(t) - \dot{x}_i(t))]. \quad (13)$$

Here $\rho_i, \gamma_i > 0$ are fixed constants. Unlike the algorithm (8), the protocol (13) requires the agents to use not only *relative measurements* $x_j - x_i$ and $\dot{x}_j - \dot{x}_i$, but also *absolute values*

of their velocities $\dot{x}_i(t)$. In the next sections second order consensus algorithms that use only relative measurements will be considered; however, unlike (13), those algorithms cannot be directly reduced to the first-order dynamics and, in general, do not provide the existence of limits (5).

To find conditions providing consensus under the algorithm (13), a useful trick was suggested in [54], introducing N “virtual agents” indexed $N+1$ through $2N$, whose “opinions” are given by $x_{j+N}(t) := x_j(t) + \gamma \dot{x}_j(t)$ (where $j = 1, \dots, N$). Assuming that $\rho_j \gamma_j > 1$ for any j , the “augmented” system of $2N$ agents obeys equations that are similar to the dynamics (8). Namely, for any $j = 1, \dots, N$ it can be shown that

$$\begin{aligned}\dot{x}_j &= \gamma_j^{-1} [x_{j+N} - x_j] \\ \dot{x}_{j+N} &= \gamma_j \sum_{k=1}^N a_{jk} [x_{k+N} - x_{j+N}] + (\rho_j - \gamma_j^{-1}) [x_j - x_{j+N}].\end{aligned}$$

The interaction graph of this extended system includes the interaction graph $\mathcal{G}[A^\top]$ of the original system (13) and N additional “virtual” nodes, connected to the “actual” nodes by pairs of arcs $(j, j+N)$, $(j+N, j)$, where $j = 1, \dots, N$. This “augmented” graph is rooted if and only if the interaction graph $\mathcal{G}[A^\top]$ is rooted. Theorem 2 implies the following.

Theorem 5: [54] Let $\rho_j \gamma_j > 1$. Then the protocol (13) establishes consensus (5) if and only if the interaction graph $\mathcal{G}[A^\top]$ is rooted; if this condition holds, the protocol also provides that $\dot{x}_j(t) \rightarrow 0$ as $t \rightarrow \infty$ for any $j = 1, \dots, N$.

Notice that the condition $\rho_j \gamma_j > 1$ is not necessary for consensus and can be relaxed (an example will be considered in Section V). Analogous results can be obtained for discrete-time second order consensus algorithms [57], [58].

IV. CONSENSUS OVER TIME-VARYING GRAPHS AND DELAY ROBUSTNESS

A substantial restriction of the French-DeGroot algorithm and its continuous-time counterpart is the fixed interaction graph: at each step an agent interacts to the same set of teammates. Many applications, however, lead to multi-agent systems with *time-varying* interaction topologies. An example of such systems is given by the *bounded confidence* models [10], [11], [31], [59] of opinion dynamics, where each person interacts only to “like-minded” individuals, whose opinions are close to his/her own opinion. Mathematically this implies that the gain w_{ij} in (2) depends on the deviation $|x_i - x_j|$ and vanishes when this deviation becomes large. In networks of mobile robots, analogous state-dependent interaction graphs arise due to *range-restricted* communication and sensing: a robot can interact only with its *nearest neighbors* and has no information about the distant teammates [60]–[63]. In the influential paper [64], which has triggered extensive studies of multi-agent consensus in systems and control theory [49], [65], [66], a model of particle ensemble was proposed, where particles align their headings, using the nearest neighbor rule. In this section we consider time-varying extensions of the algorithms (2) and (8); using these extensions, one can also generalize Theorem 5 to the case of time-varying gains $a_{jk}(t)$.

A. The Time-Varying French-DeGroot Algorithm

Given a sequence of stochastic $N \times N$ matrices $\{W(t)\}_{t=0}^\infty$, consider the following time-varying counterpart of (3)

$$x(t+1) = W(t)x(t), \quad t = 0, 1, \dots \quad (14)$$

The SIA condition (6), necessary and sufficient for consensus (5) in the time-invariant case, should be replaced by

$$\lim_{t \rightarrow \infty} [W(t)W(t-1) \dots W(0)] = \mathbb{1}_N p^\top, \quad p \in \mathbb{R}^N. \quad (15)$$

Since $W(t)$ are stochastic matrices, $p = (p_1, \dots, p_N)^\top$ is a discrete probability distribution: $p_i \geq 0$ and $p^\top \mathbb{1}_N = 1$.

Unlike the static graph case $W(t) \equiv W$, the SIA condition for each individual matrix $W(t)$, in general, *does not* imply the convergence (15). In general, for the convergence of infinite matrix products (15) or, equivalently, the regularity of heterogeneous Markov chains [45], only sufficient conditions are known. One of the general criteria, offered by Wolfowitz [44], guarantees the convergence (15) when $W(t)$ varies in some finite set of SIA matrices, such that any *finite* product of matrices from this set is also SIA. The Wolfowitz theorem allows to investigate the Vicsek model [65] and more general consensus algorithms [34]. Other criteria for the convergence of matrix products (15), based on the use of *ergodicity* or the *scrambling* coefficient have been obtained in [67]–[69] and transformed into graphical consensus criteria in [66], [70].

An alternative approach to the analysis of algorithm (14) exploits its *contraction property*. The *convex hull* spanned by the agents’ opinions, that is, the interval $J(t) = [\min_j x_j(t), \max_j x_j(t)]$, is *non-expanding*: $J(t) \supseteq J(t')$ whenever $t \leq t'$. Therefore, its diameter $\max_j x_j - \min_j x_j$ may serve as the *Lyapunov function*. This idea does not rely on the linearity of the system (14) and can be extended to a wide class of *nonlinear* consensus algorithms [49], [62], [71]–[73].

A typical assumption, adopted in consensus criteria for time-varying systems, is as follows [74].

Assumption 1: There exists $\varepsilon > 0$ such that $w_{ii}(t) \geq \varepsilon$ for any i , and all other entries $w_{ij}(t)$ belong to $\{0\} \cup [\varepsilon; 1]$.

Remark 2: Assumption 1 implies that all positive weights are *uniformly positive*, and hence the couplings among the agents cannot “loosen” over time. Even if the structure of the interaction graph remains constant, this restriction cannot be completely discarded, as demonstrated by the following example. Let the opinions of $N = 2$ agents obey the equations

$$\begin{aligned}x_1(t+1) &= (1 - w(t))x_1(t) + w(t)x_2(t) \\ x_2(t+1) &= w(t)x_1(t) + (1 - w(t))x_2(t).\end{aligned} \quad (16)$$

Then $x_1(t+1) - x_2(t+1) = (1 - 2w(t))[x_1(t) - x_2(t)] = \dots = \Pi(t)[x_1(0) - x_2(0)]$, where $\Pi(t)$ stands for the product

$$\Pi(t) = [1 - 2w(t)] \dots [1 - 2w(0)].$$

It can be shown that if $w(t) \in (0; 1/2)$ are chosen in such a way that $\sum_{t=0}^\infty w_i(t) < \infty$, then the limit $\Pi(\infty) = \lim_{t \rightarrow \infty} \Pi(t)$ exists, and $\Pi(\infty) \in (0; 1)$. In other words, if the initial opinions of the agents are different, $x_1(0) \neq x_2(0)$, this disagreement cannot be completely removed by the protocol (21) since $\lim_{t \rightarrow \infty} (x_1(t) - x_2(t)) \neq 0$.

Under Assumption 1, the standard sufficient condition for consensus is the *repeated connectivity*, which is, however, not necessary. We start with relevant definitions.

Definition 16: (Jointly rooted sequence of graphs) A sequence of graphs $\mathcal{G}_1 = (\mathcal{V}_1, \mathcal{E}_1), \dots, \mathcal{G}_m = (\mathcal{V}_m, \mathcal{E}_m)$ (with $m \leq \infty$) is *jointly rooted* if their union $\bigcup \mathcal{G}_i \triangleq (\bigcup \mathcal{V}_i, \bigcup \mathcal{E}_i)$ is rooted. For weighted graphs $\mathcal{G}[A_i]$ the joint rootedness means that the graph $\mathcal{G}[A_1 + \dots + A_m]$ is rooted.

Definition 17: (Jointly rooted time-varying graph) The time-varying interaction graph $\mathcal{G}[W(\cdot)^\top]$ of the system (14) is said to be *infinitely jointly rooted* if the infinite sequence of graphs $\{\mathcal{G}[W(t)^\top]\}_{t \geq t_0}$ is jointly rooted for any t_0 . The interaction graph is *repeatedly jointly rooted* [66] if there exists a *period* $T \geq 1$ such that the sequence of graphs $\mathcal{G}[W(t_0)^\top], \mathcal{G}[W(t_0 + 1)^\top], \dots, \mathcal{G}[W(t_0 + T - 1)^\top]$ is jointly rooted for any $t_0 \geq 0$. The infinite and repeated strong connectivity may be defined likewise.

The following consensus criterion was proved, in different formulations, in [34], [49], [66], [74].

Theorem 6: Suppose that Assumption 1 holds. If the protocol (14) establishes consensus (5), then its interaction graph is *infinitely jointly rooted*. The inverse is true for *undirected* interaction graph. In general, consensus is established if the interaction graph is *repeatedly jointly rooted*.

The infinite rootedness appears to be sufficient not only for undirected graphs but also for balanced and “cut-balanced” graphs [39], some other relaxations can be found in [74]. The protocols with undirected and balanced graphs, similar to the time-invariant case, establish *average* consensus (11) since the sum of agents’ opinions $x_1(t) + \dots + x_N(t)$ remains constant. However, for a general directed interaction graph the infinite joint rootedness is *not* sufficient, and establishing necessary and sufficient conditions for consensus remains a non-trivial open problem. An important step in this direction has been taken in the recent paper [75], considering the following problem: given a finite set of stochastic matrices $\mathcal{W}$, to check whether the protocol (14) establishes consensus for *any* sequence of matrices $W(t) \in \mathcal{W}$ or not. It was shown that this problem is *decidable* (the algorithm, answering this question in finite time exists), being however NP-hard.

Finally, it should be noticed that Theorem 6 can be extended to a wide class of nonlinear consensus algorithms [49], [71].

B. Continuous-time consensus over time-varying graphs

A time-varying counterpart of the continuous-time consensus protocol (8) is given by the following

$$\dot{x}(t) = -L[A(t)]x(t), \quad t \geq 0. \quad (17)$$

Similar to the discrete-time case, non-zero weights $a_{ij}(t)$ are usually assumed to be uniformly positive; this restriction cannot be completely discarded. Typically the matrix $A(t)$ is bounded, and hence $m > 0$ and $M > m$ exist such that

$$a_{ij}(t) \in \{0\} \cup [m, M] \quad \forall i, j = 1, \dots, N \quad \forall t \geq 0, \quad (18)$$

Usually the following *dwell-time* assumption is adopted.

Assumption 2: The matrix-valued function $A(t)$ is piecewise-constant and its consecutive switching instants

are separated by the *dwell time* $\tau_D > 0$. In other words, there exists an increasing sequence $t_1 < \dots < t_n < \dots$ such that $t_{n+1} - t_n \geq \tau_D$ and $A(t)$ is constant for $t \in [t_n, t_{n+1})$.

Assumption 2 allows to reduce the dynamics (17) to (14) via discretization [34]. Under this assumption, one can define infinite and repeated joint rootedness similar to Definition 17: the interaction graph is infinitely jointly rooted if the graphs $\{\mathcal{G}[A^\top(t)]\}_{t \geq t_0}$ are jointly rooted for any $t_0 \geq 0$ and repeatedly jointly rooted if a period $T > 0$ exists such that the iteration graphs $\{\mathcal{G}[A^\top(t)]\}_{t=t_0}^{t_0+T}$ are jointly rooted for any $t_0 \geq 0$.

Theorem 7: [34] Theorem 6 remains valid for (17), replacing Assumption 1 by (18) and Assumption 2.

The assumption of repeated joint connectivity is also referred to as the UQSC (uniform quasi-strong connectivity) condition [62]. As shown in [34], this condition (together with Assumption 2) in fact provides exponential convergence to consensus; for such a stronger version of consensus the UQSC condition is in fact necessary [62]. However, like in the discrete-time case, for consensus without exponential convergence the repeated rootedness is only sufficient but not necessary. Necessary and sufficient condition for consensus are known only for special graphs [39], [76].

The positive dwell-time assumption can in fact be relaxed, although it is often adopted to transform the continuous-time dynamics (17) into the discrete-time ones. One of the first consensus criteria, substantially relaxing this technical assumption, has been obtained in [48], [77]. Given a bounded time-varying matrix $A(\cdot)$ and numbers $t, T \geq 0$ let $\mathcal{G}_{t,T} = \mathcal{G}\left[\int_t^{t+T} A(s) ds\right]$ and $\mathcal{G}_{t,T}^\delta$, where $\delta > 0$, be the subgraph of $\mathcal{G}_{t,T}$ obtained by removing the arcs of weight less than δ .

Theorem 8: [48] The protocol (17) establishes consensus if the matrix $A(t)$ is uniformly bounded and there exist such numbers $T > 0$, $\delta > 0$ and index $k \in \{1, \dots, N\}$ that the node k is a root in the graph $\mathcal{G}_{t,T}^\delta$ for any $t \geq 0$.

Theorems 7 and 8 can be extended to some *nonlinear* consensus algorithms, providing the non-expansion of the convex hull, spanned by the agents’ states [62], [72], [73].

C. The effect of delay

In practical applications, delays in the measured and transmitted data are inevitable. Delays, in general, can deteriorate consensus and even lead to the system instability, which gives rise to the problem of consensus *robustness* against delays.

The simplest situation, which has been thoroughly studied in the literature [48], [71]–[73], is the presence of *communication delays*: the i th agent has direct access to its own state $x_i(t)$, however, the states of other agents are retarded, being e.g. sent through delaying communication channels. Denoting the delay in data, sent from the j th node to the node i , by $\tau_{ij}(t)$, the protocol (14) is replaced by the following

$$x_i(t+1) = w_{ii}(t)x_i(t) + \sum_{j \neq i} w_{ij}x_j(t - \tau_{ij}(t)) \quad \forall i. \quad (19)$$

Analogy of the protocol (17) with communication delays is

$$\dot{x}_i(t) = \sum_{j=1}^N a_{ij}(t)[x_j(t - \tau_{ij}(t)) - x_i(t)] \quad \forall i. \quad (20)$$

It can be shown that *bounded* communication delay *does not change* the convergence properties of the protocol: if $\sup_{t \geq 0} \tau_{ij}(t) < \infty$ for any i, j then the result of Theorem 6 (respectively, Theorem 7) remains valid for the protocol (19) (respectively, (20)). The key idea of the proof [48], [71]–[73] is a modified contraction property of the algorithm: the convex hull of all agents' states over the period τ_* is non-expanding; in particular, the solutions of (19) and (20) remain bounded. Consensus thus proves to be *robust* against time-varying bounded communication delays. Furthermore, consensus is robust even against special *unbounded* communication delays [78]. At the same time, consensus values $\lim_{t \rightarrow \infty} x_i(t)$ for delayed and undelayed algorithms are different; the problem of finding consensus value of (20) is addressed in [79].

At the same time, the robustness of consensus algorithms against *self-delays* remains a challenging problem that has been solved only in a few special situations. Self-delays may be caused by some internal processes in the agents, such as the dynamics of actuators or the operator's response lag in a human-machine system. An example of such “input” or “inner” delay [80]–[82] is the driver reaction time in microscopic traffic flow models [29], [30]. Self-delays can also be introduced by sensors, measuring agents' states $x_i(t)$ or their deviations $x_j(t) - x_i(t)$. For instance, hydroacoustic sensors, measuring relative positions $x_j(t) - x_i(t)$ of underwater robots, introduce inevitable *propagation delays*, depending on the distances $|x_j - x_i|$ and the speed of sound in water.

In general, under sufficiently large self-delay the multi-agent system becomes unstable, as demonstrated by the following simple example. Consider the system of $N = 2$ agents, applying the following protocol with delay

$$\begin{aligned} \dot{x}_1(t) &= a(x_2(t - \tau) - x_1(t - \tau)) \\ \dot{x}_2(t) &= a(x_1(t - \tau) - x_2(t - \tau)). \end{aligned} \quad (21)$$

Here $a > 0$ is a constant gain and $\tau > 0$ is a delay. The deviation of the agents' states $\delta(t) \triangleq x_2(t) - x_1(t)$ obeys

$$\dot{\delta}(t) = -a\delta(t - \tau).$$

It is known [83] that the system (21) is globally asymptotically stable if $\tau < \pi/(2a)$ and unstable for $\tau(t) > \pi/(2a)$ (in the critical case $\tau = \pi/(2a)$ a periodic solution exists). In the case of time-varying delay $\tau = \tau(t)$ the stability can be proved only for $\tau(t) < 3/(2a)$ (and this delay bound is tight [83]). This example demonstrates that consensus (5) is robust only against *sufficiently small* self-delays, giving thus rise to the problem of finding the “critical” self-delay margin, below which consensus is guaranteed. Tight estimates of the critical delay are known only for time-invariant consensus protocols [51], [80], [84]–[86]. An elegant result from [51] establishes a criterion for consensus in the network

$$\dot{x}_i(t) = \sum_{j=1}^N a_{ij}[x_j(t - \tau) - x_i(t - \tau)], \quad (22)$$

where $a_{ij} = a_{ji}$ are constant and the delay $\tau > 0$ is fixed. The protocol (22) establishes consensus if and only if the

undirected interaction graph $\mathcal{G}[A]$ is connected and

$$\tau < \frac{\pi}{2\lambda_{\max}}, \quad (23)$$

where $\lambda_{\max}$ is the maximal eigenvalue of the symmetric Laplacian matrix $L[A]$. More general criteria, providing consensus in presence of heterogeneous delays, have been established in [84]–[86]. In the discrete-time case, an estimate for the critical delay margin has been offered in [80]. Consider a protocol with both communication delays and self-delays

$$x_i(t+1) = x_i(t) + \sum_{j=1}^N a_{ij}[x_j(t - \tau_{ij} - D_i) - x_i(t - D_i)]. \quad (24)$$

Here $D_i \geq 0$ stands for the constant *input delay* of agent i . The protocol (24) provides consensus if the interaction graph $\mathcal{G}[A^\top]$ is rooted and the self-delays are sufficiently small

$$(2D_i + 1) \sum_{j=1}^N a_{ij} < 1 \quad \forall i = 1, \dots, N. \quad (25)$$

Notice that (25) does not employ *communication delays* τ_{ij} , which are assumed to be fixed but may be arbitrarily large. If there are no communication delays $\tau_{ij} = 0$ and the graph is undirected $a_{ij} = a_{ji}$, the sufficient condition (25) can be tightened to the *necessary and sufficient* condition

$$\sum_{j=1}^N a_{ij} < \sin \frac{\pi}{2(2D_i + 1)} \quad \forall i = 1, \dots, N. \quad (26)$$

For time-varying interaction graphs only sufficient estimates of the delay margin are known. Most of them employ high-dimensional systems of linear matrix inequalities (LMI) [87], [88]. For undirected time-varying graphs sufficient conditions for consensus under self-delays have been obtained in [82], [89]–[91]; these criteria are valid also for nonlinear sector-bounded couplings and have applications in traffic flow models with delayed drivers' reactions [30].

V. CONSENSUS OF GENERAL DYNAMIC AGENTS

The consensus protocols (2) and (8), as well as their time-varying analogs, assume the simplest dynamics of agents. For instance, the algorithm (2) implicitly assumes the possibility to control the states $x_i(t)$ or their increments $x_i(t+1) - x_i(t)$, and the algorithm (8) uses the derivatives $\dot{x}_j(t)$ as the control inputs. In this section, we consider consensus algorithms for agents with general dynamics, which are described by state-space models, typical in control theory.

A. Consensus of homogeneous linear MIMO agents

In this section, we consider a team of N identical agents, described by the state-space MIMO model

$$\dot{x}_i(t) = Px_i(t) + Qu_i(t), \quad y_i(t) = Rx_i(t). \quad (27)$$

Here $x_j(t) \in \mathbb{R}^n$ stands for the *state vector* of the agent, $u_j(t) \in \mathbb{R}^m$ is its *control* or *input*, and $y_j(t) \in \mathbb{R}^l$ is some *output*; P , Q and R stand for some constant matrices of the appropriate dimensions. The state vectors are usually unavailable for measurement, the agents measure their own

outputs y_i and also receive outputs of their neighbors in some communication graph. In the more interesting case the agents are “non-introspective” [92] and measure only the deviations between their own and the neighbors’ outputs $y_j - y_i$.

A natural extension of the protocol (8) is as follows

$$u_i(t) = \sum_{j=1}^N a_{ij} K(y_j(t) - y_i(t)), \quad (28)$$

where K is a constant $m \times l$ matrix. The case of first-order integrator agents $\dot{x}_i = u_i$ (that is $P = 0$, $Q = R = I_n$) and $K = I_n$ corresponds to the first-order consensus algorithm (8). Unlike (8), the protocol (28) typically does not provide the existence of limits (5). For general agents (27), consensus is typically defined as asymptotic synchronization of the state vectors. We now give a formal definition of consensus.

Definition 18: The protocol (28) establishes *state consensus* (or synchronization) among the agents (27) if

$$|x_i(t) - x_j(t)| \xrightarrow[t \rightarrow \infty]{} 0 \quad \forall i, j = 1, \dots, N, \quad (29)$$

for any initial condition $x_1(0), \dots, x_N(0)$.

The following fundamental consensus criterion has been established in [53], [93]–[95], giving criteria for state consensus among the agents (27).

Theorem 9: The protocol (28) establishes state consensus among the agents (27) if and only if two conditions hold:

- 1) the interaction graph $\mathcal{G}[A^\top]$ is rooted;
- 2) for non-zero eigenvalues $\lambda_2, \dots, \lambda_N$ of the Laplacian (1), the matrices $P - \lambda_j Q K R$ are Hurwitz (that is, the feedbacks $u_j = -\lambda_j K y_j$ stabilize the agent).

If 1) and 2) hold then the convergence in (29) is exponential.

Theorem 9 can be reformulated for discrete-time agents

$$x_i(t+1) = P x_i(t) + Q u_i(t), \quad y_i(t) = R x_i(t), \quad (30)$$

coupled via the same protocol (28), replacing the word “Hurwitz” with “Schur stable”. A more important extension is concerned with *dynamic* controllers [93], [94]

$$\dot{z}_i = F z_i + G \xi_i, \quad \xi_i = \sum_{j=1}^N a_{ij} (y_j - y_i), \quad (31)$$

$$u_i = H z_i + L \xi_i.$$

The control algorithm (31) is determined by constant matrices F, G, H, L of the appropriate dimensions.

Theorem 10: The protocol (31) establishes consensus among the agents (27) if and only if two conditions hold:

- 1) the interaction graph $\mathcal{G}[A^\top]$ is rooted;
- 2) for any non-zero eigenvalue $\lambda = \lambda_2, \dots, \lambda_N$ of the Laplacian (1), the dynamic controller

$$\begin{aligned} \dot{z}_i &= F z_i - \lambda G y_i \\ u_i &= H z_i - \lambda L y_i. \end{aligned} \quad (32)$$

stabilizes the agent (27).

Theorems 9-10 reduce consensus to the simultaneous stabilization, enabling one not only to solve the analysis problem, but also to *design* consensus algorithm (28) or (31) for given agents and a fixed interaction graph [92], [94], [96]–[98]. Using adaptive control techniques, a “fully distributed”

consensus algorithm can be designed, which does not use any information about the interaction graph (such as e.g. the spectrum λ_i) [98]. These design methods can also be extended to Lur’e systems with slope-restricted nonlinearities [98] and some classes of heterogeneous agents [99].

Besides the state-space approach, used in Theorems 9-10, consensus algorithms for general agents (27) can be examined by various frequency-domain methods [85], [86], [100], [101].

Example 1. Consider a group of agents of dynamic agents

$$\ddot{x}_i + \varrho \dot{x}_i = u_i, \quad y_i = x_i + \gamma \dot{x}_i, \quad i = 1, \dots, N. \quad (33)$$

applying the protocol (28) with $K = 1$. Obviously, the resulting multi-agent system obeys (13) with identical coefficients $\rho_i = \rho$ and $\gamma_i = \gamma$. Suppose that the undirected interaction graph $\mathcal{G}[A^\top]$ is *undirected* ($a_{ij} = a_{ji}$) and connected. Then the eigenvalues $\lambda_2, \dots, \lambda_N$ are *real* and positive, and hence the controller $u_i = -\lambda u_i$ stabilizes the agent (34) for any $\lambda = \lambda_2, \dots, \lambda_N$. Theorem 9 now implies that for any $\rho, \gamma > 0$ the protocol establishes consensus (29). Since the convergence in (29) is exponential, in fact consensus in the stronger sense (5) also holds and $\dot{x}_i \rightarrow 0$. Thus Theorem 5 in general gives only *sufficient* but not necessary condition for consensus: for undirected graphs and homogeneous coefficients ϱ_i, γ_i the condition $\varrho_i \gamma_i > 1$ can be discarded.

Example 2. Consider now a team of agents

$$\ddot{x}_i = u_i, \quad y_i = \alpha x_i + \beta \dot{x}_i, \quad i = 1, \dots, N. \quad (34)$$

applying the protocol (28) with $K = 1$. Here α, β are constant scalar coefficients. The resulting multi-agent system obeys then the following second order equations

$$\ddot{x}_i = \sum_{j=1}^N a_{ij} [\alpha(x_j(t) - x_i(t)) + \beta(\dot{x}_j(t) - \dot{x}_i(t))] \quad \forall i. \quad (35)$$

Theorem 9 guarantees consensus $x_i - x_j \rightarrow 0$, $\dot{x}_i - \dot{x}_j \rightarrow 0$, when $\alpha, \beta > 0$ and the interaction graph is *undirected* and connected. For directed graphs, however, an additional restriction on α and β arises, involving the real and imaginary parts of the eigenvalues λ_j [34], [102].

Second-order consensus algorithms, analogous to (13) and (35), have attracted substantial attention because of numerous applications to control of multi-vehicle formations. In particular, the effects of delays [102]–[104], noises [105] and time-varying interaction graphs [34], [103], [106] have been thoroughly studied. Besides linear algorithms (13), (35), consensus algorithms with nonlinear couplings among the agents have been also examined [35], [107]–[109].

B. Master Stability Function (MSF) approach

Although rigorous proof of Theorem 9 appeared only in [93], its *non-stationary* extension, has been widely used in the physical literature under the name of the *Master Stability Function* (MSF) method [110]. This method allows to get a *local* synchronization condition for nonlinear networks

$$\dot{x}_i(t) = f(x_i(t)) + \sum_{j=1}^N a_{ij} [g(x_j(t)) - g(x_i(t))], \quad i \in 1 : N, \quad (36)$$

in a small vicinity of an equilibrium or periodic solution $x_1(t) = \dots = x_N(t) = x_*(t)$. Here $f, g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ stand for some C^1 -smooth mappings (in general, nonlinear).

Linearizing the system at the solution $x_*(t)$, one arrives at the linear system with time-varying coefficients that can be considered as a team of agents

$$\dot{x}_i(t) = P(t)x_i(t) + Q(t)u_i(t), \quad y_j(t) = R(t)x_j(t), \quad (37)$$

applying non-stationary control protocol

$$u_j(t) = \sum_{k=1}^N a_{jk} K(t)(y_k(t) - y_j(t)). \quad (38)$$

Notice that in general, the asymptotic stability of a time-varying linearization does not imply the stability of a solution. The stability can be proved however under additional *regularity* assumption [111], which holds for linearizations at limit cycles.

For any $z \in \mathbb{C}$ consider the system of differential equations

$$\dot{\xi}(t) = [P(t) - zQ(t)K(t)R(t)]\xi(t),$$

and let $\lambda_{max}(z)$ stand for its maximal Lyapunov exponent [111]. The function λ_{max} is referred to as the *master stability function* (MSF) of the system. In the stationary case $\lambda_{max}(z)$ is simply the maximal real part of the eigenvalues of the matrix $P - zQKR$ and hence $\lambda_{max}(z) < 0$ if and only if $P - zQKR$ is Hurwitz.

The MSF approach [110] gives an extension of Theorem 9 to time-varying agents (37) and protocol (38).

Theorem 11: The protocol (38) establishes consensus (29) among the agents (37) if and only if two conditions hold:

- 1) the interaction graph is rooted;
- 2) for the eigenvalues $\lambda_2, \dots, \lambda_N$ of the Laplacian $L[A]$ one has $\lambda_{max}(\lambda_j) < 0$.

Applying Theorem 11 to the Jacobian matrices $P(t) = Df(x_*(t))$, $Q(t) = Dg(x_*(t))$ and $R(t) = K(t) = I$, one can easily transform it into the criterion of *local* state synchronization in the nonlinear networked system (36).

C. Nonlinear multi-agent systems

A natural extension of multi-agent systems, addressed in the foregoing, is a system where the agents

$$\begin{aligned} \dot{x}_i(t) &= f(x_i(t), u_i(t)) \\ y_i(t) &= h(x_i(t)) \end{aligned}, \quad i = 1, \dots, N, \quad (39)$$

are nonlinear and coupled via nonlinear algorithm

$$u_j(t) = \sum_{k=1}^N a_{jk}(t) \varphi_{jk}(y_k(t) - y_j(t)). \quad (40)$$

Here $\varphi_{jk}(\cdot)$ are mappings, referred to as *coupling maps* or *couplings*. The protocols (17) and (28) employ *linear* couplings $\varphi_{jk}(y) = y$, however, nonlinear coupling maps naturally arise in many applications, such as oscillator networks [72], [112]–[114] and coordination with range-restricted communication [61]–[63]. In practice linear protocols may become nonlinear due to nonlinear distortions in measurements.

To establish the condition for consensus (29) in the non-linear multi-agent system (39),(40) is a non-trivial and in fact open problem. The MSF approach, discussed in the previous subsection, allows to establish *local* consensus conditions, provided that the interaction graph is fixed $a_{jk} = \text{const}$. Other linearization-based methods, applicable to time-varying graphs, have been proposed in [115], [116]. These “local” methods, however, do not provide constructive estimates for the basin of attraction and lead to non-trivial computational procedures. For instance, the MSF computation requires to estimate the Lyapunov exponents of non-stationary systems.

The results, establishing *global* consensus (29) in a general system (39),(40), are very limited. Many of them employ the idea of contraction [117] or, more generally, *incremental* Lyapunov functions $\bar{V}(X) = \sum_{i,j=1}^N V(x_j - x_i)$, where V is a positive definite function and $X = (x_1, \dots, x_N)$ stands for the joint state of the system. A basic idea of designing this Lyapunov function is to find an *incremental* constraint

$$\sum_{j,k=1}^N F(u_j - u_k, y_j - y_k) \geq 0, \quad (41)$$

where $F(u, y)$ is some known function, e.g. a quadratic form.

To illustrate one of the possible constraints (41), suppose that $a_{ij} = a_{ji} \geq 0$ and the couplings are skew-symmetric $\varphi_{jk}(y) = -\varphi_{kj}(y)$, satisfying thus a counterpart of Newton’s Third Law. Assume also that the couplings are “passive” in the sense that $\varphi_{jk}(y)^\top y \geq 0$. A straightforward computation [107], [108], [118] allows to prove that $\sum_j u_j = 0$ and

$$0 \leq 2 \sum_j u_j^\top y_j = N^{-1} \sum_{j,k} (u_j - u_k)^\top (y_j - y_k).$$

Constraints (41) often entail the existence of incremental Lyapunov functions $\bar{V}(X)$. To guarantee that $\frac{d}{dt} \bar{V}(X) = \sum_{i,j=1}^N \frac{d}{dt} V(x_j - x_i) \leq 0$, it suffices to prove that

$$\dot{V}(x_1(t) - x_2(t)) + F(u_1(t) - u_2(t), y_1(t) - y_2(t)) \leq 0 \quad (42)$$

for any two solutions x_1, u_1, y_1 and x_2, u_2, y_2 of (39). The condition (42) is the so-called *incremental dissipativity* [119] with the (incremental) storage function V . This condition, extending the conventional Willems dissipativity [120], proves to be a convenient tool in consensus analysis [121], [122].

The idea of incremental Lyapunov functions has been widely used for consensus (synchronization) of biological [119], [123] and physical oscillators [124]–[127]. Such functions also allow to get “multi-agent” counterparts of the circle and the Popov criteria stability criteria, providing consensus among *linear* agents (27), coupled via *nonlinear* protocols (40), see [90], [107]–[109]. Analogous incremental Lyapunov functions can be applied to prove consensus between discrete-time agents [128].

The method of incremental Lyapunov functions assumes that agents are identical (homogeneous). This assumption can be discarded in the situation where agent are *passive* systems

$$\begin{aligned} \dot{x}_i(t) &= f_i(x_i(t), u_i(t)) \\ y_i(t) &= h_i(x_i(t)) \end{aligned} \quad (43)$$

By definition, passivity means that $\dim u_i = \dim y_i$ and a storage function $V_i(x_i)$ exists such that

$$V_i(x_i(t)) - V_i(x_i(0)) \leq \int_0^t y(s)^\top u(s) ds.$$

Suppose that the agents are coupled by a distributed algorithm

$$u_j(t) = \sum_{k=1}^N a_{jk}(y_k(t) - y_j(t)). \quad (44)$$

It can be shown that if the individual storage functions $V_i(x_i)$ are positive definite and radially unbounded, the protocol (44) established a “partial” consensus with respect to the outputs.

Definition 19: The protocol (28) establishes *output* consensus (or synchronization) among the agents (27) if

$$|y_i(t) - y_j(t)| \xrightarrow[t \rightarrow \infty]{} 0 \quad \forall i, j = 1, \dots, N. \quad (45)$$

for any initial condition $x_1(0), \dots, x_N(0)$.

The following elegant result [129, Theorem 8.5] provides a criterion for consensus in the system (43), (44).

Theorem 12: Suppose that V_i are positive definite and radially unbounded, and the interaction graph $\mathcal{G}[A^\top]$ is *strongly connected*. The the protocol (44) establishes output consensus (45) between the agents (43).

Other consensus criteria for passive agents, dealing with time-varying interaction graphs, nonlinear couplings and delays, can be found in [107], [108], [118], [129]–[131]. Besides passive agents, the problems of analyzing and design consensus algorithms have been thoroughly studied for *feedback passive* or “passifiable” agents, which can be rendered passive by some output feedback. Such agents can be synchronized by sufficiently strong couplings, and the respective coupling gains can be adapted by using standard model reference adaptive control (MRAC) approaches [132], [133].

VI. APPLICATIONS OF CONSENSUS PROTOCOLS

Besides the aforementioned models of social dynamics and collective decision making, consensus protocols from previous sections have found many applications in multi-agent control, signal processing, and distributed computing. Numerous applications can be found in recent surveys and monographs [34]–[36], [53], [129], [134], [135] and references therein. In this section we discuss only a few of these applications.

A. Formation control of mobile agents

Consider a group of mobile autonomous agents $\dot{q}_i = u_i \in \mathbb{R}^d$, where $q_i(t)$ stands for the position of i th agent and $u_i(t)$ is its velocity, being also the *control input*. Each agent can measure its position relative to the “nearest” neighbors; the neighboring relations (in general, asymmetric) are described by the arcs of a fixed interaction graph $\mathcal{G}[A^\top]$: when $a_{ij} > 0$, the i th agent is able to measure the vector $q_j(t) - q_i(t)$ at any time. The agents cooperate to create the desired *formation* in the space. This goal may be formally defined as follows: for a given set of points $e_1, \dots, e_N \in \mathbb{R}^d$, the i th agent has to converge to the points $e_i + \delta$, where $\delta \in \mathbb{R}^n$ is a common

shift vector; in particular, the agents’ relative positions $q_j(t) - q_i(t) \xrightarrow[t \rightarrow \infty]{} \varepsilon_{ij} \triangleq e_j - e_i$.

Assuming that each agent knows the desired positions relative to its neighbors, consider the following control algorithm

$$\dot{q}_i(t) = u_i(t) = \sum_{j=1}^N a_{ij}[q_j(t) - q_i(t) - \varepsilon_{ij}]. \quad (46)$$

Denoting $x_i(t) \triangleq q_i(t) - e_i$, the algorithm (46) shapes into (8). Applying Theorem 2, one shows that the existence of a root in $\mathcal{G}[A^\top]$ implies the convergence $x_i(t) - e_i \rightarrow \delta \quad \forall i$, where δ depends on the initial condition.

The same idea can be applied to formations of agents with higher order dynamics, where the agents do not converge to some points, but move, keeping the desired shape [136]. More sophisticated algorithms for formation control of mobile robots and vehicles can be found in [34], [35], [60], [63], [129], [131], [135]. Many of these algorithms employ *potential functions*, penalizing the distances between the agents and deviation of their velocities; the gradient descent procedure, minimizing the potential function, gives then a distributed algorithm for formation control.

B. Rendezvous problem

In the special case where $e_1 = \dots = e_N$ (the desired shape is a singleton), the algorithm (46) coincides with (8) and provides *rendezvous* of the mobile agents $\dot{q}_i = u_i$, that is, the agents gather at some point, depending on their initial positions. A simple modification of (46) allows to solve the rendezvous problem for more realistic *double integrator* agents $\ddot{x}_i = u_i$, where the control input stands for the agent’s acceleration. Assume that agents are able to measure their positions and velocities relative to the “neighbors”, and also have access to their *absolute* velocities $\dot{q}_j$ (e.g. ground vehicles, equipped with speed meters, or marine crafts, using speed logs). Then the algorithm (13) provides their rendezvous, assuming that the conditions of Theorem 5 hold.

It should be noticed, however, that the interactions between realistic mobile robots are *range restricted*: each agent has a bounded visibility area and is able to observe only “nearest” agents, being inside this area. So the practical rendezvous algorithms should also *maintain connectivity* of the network, being a prerequisite for agents’ gathering at one point. A number of rendezvous algorithms for networks with range-restricted interactions have been proposed in [61]–[63], [137].

C. Distributed parameter estimation and filtering

Algorithms for *average consensus* can be efficiently used for estimation in distributed sensor networks. Consider a network of identical N sensors (e.g. smoke detectors), distributed over some area and measuring some slowly varying physical quantity. Assuming that the actual quantity is equal to $a \in \mathbb{R}$, the sensors’ displays are independent identically distributed (i.i.d.) Gaussian random variables $x_n \sim \mathcal{N}(a, \sigma^2)$, where σ^2 is the sensor’s variance. Each sensor communicates to a few neighbors; the graph of interconnections is assumed to be undirected and static. Using the algorithm (2), where $W = W^\top$

is a weighted adjacency matrix for the communication graph, the sensors are able to compute the average measurement

$$\bar{x} = \frac{x_1 + \dots + x_N}{N} \sim \mathcal{N}(a, \sigma^2/N),$$

whose variance is much less than the individual variances. A similar problem is *distributed inference* [138], where the sensors have to distinguish between two hypotheses and average their log-likelihood ratios [139].

An important problem, regarding the distributed averaging procedures, is the optimal choice of the adjacency matrix W , providing the fastest convergence [139]–[141].

Similar ideas can be applied to the estimation of dynamic processes by distributed sensor networks [142]–[145]. Combining the usual Kalman filter and a consensus protocol, averaging the observation from different sensors enables one to estimate the state of a dynamic system with high accuracy.

D. Constrained consensus and distributed optimization

Consensus-based algorithms, discussed in the previous sections, can be modified in order to bring the agents' states to some desired set, which is known by none or only a few of them. An example of such a protocol is a *containment* control algorithm [35], already mentioned in Section III-D, where the desired set is a polytope, spanned by several independent leaders. A more general problem is *aggregation* [146] of the agents at a given convex closed set ("target"). Unlike containment control algorithms, where the desired set may be available to none of the agents (unless some agent is connected to all of the leaders), in the target aggregation problem some "informed" agents are not only aware of the desired set, but also able to calculate the projection of any point onto it. In this section we consider extensions of the aggregation consensus protocols from [146], proposed in [26] and [27] and providing convergence of the agents' states to a common point in an implicitly defined convex set.

Consider N agents with vector state vectors $x_i(t) \in \mathbb{R}^d$. The common goal of agents is to converge to some point, which belongs to the intersection of sets $X = \bigcap_{i=1}^N X_i$. Each of the sets X_i is *convex and closed*, and their intersection is non-empty $X \neq \emptyset$. We assume that the i th agent is aware only of the corresponding set X_i , but not of the other sets X_j (where $j \neq i$). Furthermore, agent i is able to compute the projection onto its set X_i ; the corresponding projection operator is denoted by $P_i : \mathbb{R}^d \rightarrow X_i$; that is, $P_i(x) \in X_i$ is the point, closest to x in the Euclidean distance. Notice that some of the sets X_i can coincide with the whole space $\mathbb{R}^d$, in which case one has $P_i(x) = x$.

The following discrete-time consensus-type algorithm, providing the convergence of the agents' states to a point in X , have been proposed in [26]

$$x_i(t+1) = P_i \left(\sum_{j=1}^N a_{ij} x_j(t) \right). \quad (47)$$

A continuous-time analog of the algorithm (47), examined in a more recent paper [27], is as follows

$$\dot{x}_i(t) = \sum_{j=1}^N a_{ij} [x_j(t) - x_i(t)] + P_i(x_i(t)) - x_i(t). \quad (48)$$

Assuming that the graph $G[A^\top]$ is *strongly* connected (which assumption cannot be replaced by rootedness), it is possible to prove that $x_i(t) \xrightarrow[t \rightarrow \infty]{} X$ (more rigorously, $\inf_{x \in X} |x_i(t) - x| \rightarrow 0$). This implies that the algorithms (47) and (48) can be rewritten, respectively, as

$$\begin{aligned} x_i(t+1) &= \sum_{j=1}^N a_{ij} x_j(t) + \delta_i(t), \\ \dot{x}_i(t) &= \sum_{j=1}^N a_{ij} [x_j(t) - x_i(t)] + \delta_i(t), \end{aligned}$$

where the "projection errors" are decaying $\delta_i(t) \rightarrow 0$. Since the convergence to consensus is exponential, it is robust to these decaying "disturbances", and hence the agents' states $x_i(t)$ converge to a common point $x^* \in X$. This result retains its validity for time-varying *repeatedly strongly connected* graphs [26], [27].

Assuming that X_i is the set of minima of some convex function, gathering the agents' states into their intersection X can be considered as a *convex optimization* problem [27]. Further relations between the "constrained consensus" problem and distributed optimization can be found in [26].

CONCLUSIONS

This paper offers a brief introduction into a rapidly developing theory of multi-agent consensus protocols. We give a review of main historical milestones and consider basic algorithms, providing consensus between general dynamical agents, together with mathematical techniques for their analysis. We also discuss some applications of consensus protocols.

The number of applications, employing consensus protocols and similar distributed policies, is constantly growing, motivating the rapid development of consensus theory and relevant mathematical techniques. In this article, we intentionally confine ourselves to the results and concepts that have already become mature and are used most often. Our review hence does not cover many "cutting edge" problems and applications, related to mobile sensor networks [147], [148], consensus in quantum systems [149], PageRank and web aggregation [150] and networks with antagonistic interactions [18], [19], [151]. Many technical results, which require advanced mathematical and control-theoretic tools, are not covered in our review either; among them are protocols with asynchronous state updates [25], [70], [152], consensus over stochastic networks [153], [154], event-triggered consensus protocols [155], [156], quantization effects and discontinuous consensus dynamics [157]. Some of these problems are discussed in the companion article [158], devoted to a closely related topic of consensus in networks.

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